

Issued: 8/2025

PSP-Based MOSVAR v1.4.0

Arizona State University:

G. Gildenblat, Z. Zhu, W. Wu, and X. Li

Sentinel IC Technologies: J. Victory

TowerJazz Semiconductor: Z. Yan

Freescale Semiconductor: C. McAndrew

Technical contributions of G. Coram (Analog Devices) and R. Poore (Agilent) are gratefully acknowledged.

Copyright ©2011 Arizona State University and Freescale Semiconductor. All Rights Reserved.

Copyright ©2025 Silicon Integration Initiative, Inc. All Rights Reserved

Varactor model developers

- Arizona State University
- TowerJazz Semiconductor
- CMC varactor model subcommittee

Contact information **Geoffrey Coram, Si2/CMC MOSVAR subcommittee chairman**

Software is distributed as is, completely without expressed or implied warranty or service support. ASU, Si2, their members and employees are not liable for the condition or performance of the software.

ASU and Si2 own the copyright and grants users a perpetual, irrevocable, worldwide, non-exclusive, royalty-free license with respect to the software as set forth below.

ASU and Si2 hereby disclaim all implied warranties.

ASU and Si2 grant the users the right to modify, copy, and redistribute the software and documentation, both within the user's organization and externally, subject to the following restrictions:

1. The users agree not to charge for the code itself but may charge for additions, extensions, or support.
2. In any product based on the software, the users agree to acknowledge developers of the software. This acknowledgment shall appear in the product documentation.
3. The users agree to obey all government restrictions governing redistribution or export of the software.
4. The users agree to reproduce any copyright notice which appears on the software and documentation on any copy or modification of such made available to others.

Title: PSP-Based MOSVAR v1.4.0

Keywords: MOS varactor; PSP model; Compact modeling; Circuit simulation

Abstract: The MOS varactor compact model is based in part on the PSP MOSFET model and is intended for analog and RF design. It includes dynamic inversion, finite poly doping, quantum mechanics, tunneling currents, and parasitics to model advanced MOS technologies. This manual contains a description of the PSP-based varactor model, including an introduction, parameter sets, model equations, and a parameter extraction procedure.

History of model

Dec 2007 Release of MOSVAR v1.0.0. The Verilog-A code was released under an open source license.

Dec 2008 Release of MOSVAR v1.1.0.

- The upper bound of resistor “rgsal” was changed from $10\ \Omega$ to $1000\ \Omega$.
- Added a warning message, controlled by parameter IGMAX, to report extraordinarily large gate tunneling current; it notes possible extraction issues with gate tunneling current and resistance parameters.
- Five redundant model parameters, namely IGCEVLW, IGOVEVW, GCOEVO, GC2EVO and GC3EVO, were removed.
- The smoothing function used in the accumulation resistance (line no. 644 in MOSVAR v1.0.0) is corrected. It should be MINA instead of MAXA.

Sept 2013 Release of MOSVAR v1.2.0.

- Added parameter EPSROXO for relative dielectric permittivity.
- Revised gate current computations to avoid numerical problems and to use GC3HVO for HVB current.
- Added parameter RACNOISE to select noise equations for Rac.
- Added more descriptive noise names.
- Added shot noise for gate currents.

July 2015 Release of MOSVAR v1.3.0.

- Extended range of parameters TOXO and NSUBO.
- Replaced $V(b_cf)$ with $V(b_ov)$ for Igov calculation.
- Replaced $V(b_cf)$ with $V(b_ci)$ for IgC calculation.

Aug 2025 Release of MOSVAR v1.4.0.

- Changed IGMAX warning to check only dc current.
- Added SWQINV to enable switching off Qinv calculation.
- Added aliases DTEMP (for DTA) and TREF (for TR).
- Updated macros ‘MINA and ‘MAXA.

History of documentation

Dec 2007 First release of user’s manual.

Dec 2008 The user’s manual was adapted to MOSVAR v1.1.0.

Sept 2013 The user’s manual was updated to MOSVAR v1.2.0.

Nov 2013 The user’s manual was updated to include gate current shot noise, which was already in the code.

July 2015 The user’s manual was updated to MOSVAR v1.3.0.

Aug 2025 The user’s manual was updated to MOSVAR v1.4.0.

Contents

	v
1 Introduction	1
1.1 Device Structure	1
1.2 An RC Circuit Model for Inversion Charge	3
1.3 Differences from the PSP MOSFET Model	3
2 Constants and Parameters	4
2.1 Circuit Simulator Variables	4
2.2 Model Constants	4
2.3 Instance Parameters	5
2.4 Special Model Parameters	5
2.5 Model Parameters	6
3 Parameter Initialization	9
3.1 General Parameters	9
3.2 Oxide Capacitance and Body Factors	9
3.3 Temperature-Related Parameters	10
3.4 Parameters Related to Polysilicon and Overlap Regions	10
3.5 Fringing Capacitance	11
3.6 Resistances	12
3.7 Noise Parameter	12
3.8 Gate Tunneling Parameters	13
4 Model Equations	15
4.1 Static Evaluations	15
4.2 Calculation of the Surface Potential at the Channel Side	16

4.3	Calculation of the Surface Potential in the Overlap Regions	18
4.4	Surface Potential without Poly Effect	19
4.5	Surface Potential with Poly Effect	19
4.6	Static Inversion Charge Calculations	19
4.7	Time-Dependent Silicon Surface Potential Calculation without Poly Effect	20
4.8	Time-Dependent Poly Surface Potential Calculation Correction	21
4.9	Quantum Mechanical Corrections	21
4.10	Accumulation Resistance Bias Dependence	22
4.11	Gate Tunneling Current Macro	22
4.12	Gate Tunneling Current	23
4.13	Currents	25
4.13.1	DC Currents	25
4.13.2	Resistor Currents	25
4.14	Terminal Charges	25
4.15	Noise	25
4.15.1	Thermal Noise	25
4.15.2	Shot Noise	26
5	Parasitic Elements and Parameter Extraction	27
5.1	Parasitic Elements	27
5.1.1	Fringing Capacitance C_{fr}	27
5.1.2	Well and Accumulation Resistance R_{ac} , R_{sub} , and R_{end}	27
5.1.3	Poly Gate Resistance R_{gsal} and R_{gpv}	28
5.2	Model Parameter Extraction	28
5.2.1	Capacitance-Related Model Parameter Extraction	28
5.2.2	Resistance-Related Model Parameter Extraction	31
5.3	Model Validation	32
A	Auxiliary Equations	35
	References	37

Section 1

Introduction

The MOSVAR varactor model is a compact model intended for analog and radio-frequency circuit design. It includes dynamic inversion, finite poly doping, quantum mechanics, gate tunneling for different polarity combinations, and parasitics to model advanced MOS technologies. It is based on the PSP model for MOS devices [1, 2]. The MOSVAR model is maintained by the Compact Model Coalition (CMC) of the Silicon Integration Initiative (Si2). The source code of MOSVAR and the most recent version of this documentation is available from the Si2 website [3].

1.1 Device Structure

Figure 1.1 shows a cross section of the standard MOS varactor offered in current MOS technologies with its equivalent circuit model superimposed on it. **g**, **bi** and **b** are the external terminals while **gii**, **gi** and **ci** are the internal nodes. See Table 1.1 for the meaning of other symbols.

Table 1.1: Components of varactor model

No.	Name	Description
1	C	Gate-channel capacitance
2	C _{fr}	Fringe and overlap capacitance
3	I _{gc}	Gate-channel current
4	I _{gov}	Gate-overlap current
5	R _{gsal}	Metal resistance
6	R _{gpv}	Poly gate resistance
7	R _{ac}	Accumulation resistance
8	R _{sub}	Substrate (in well region) resistance
9	R _{end}	End resistance

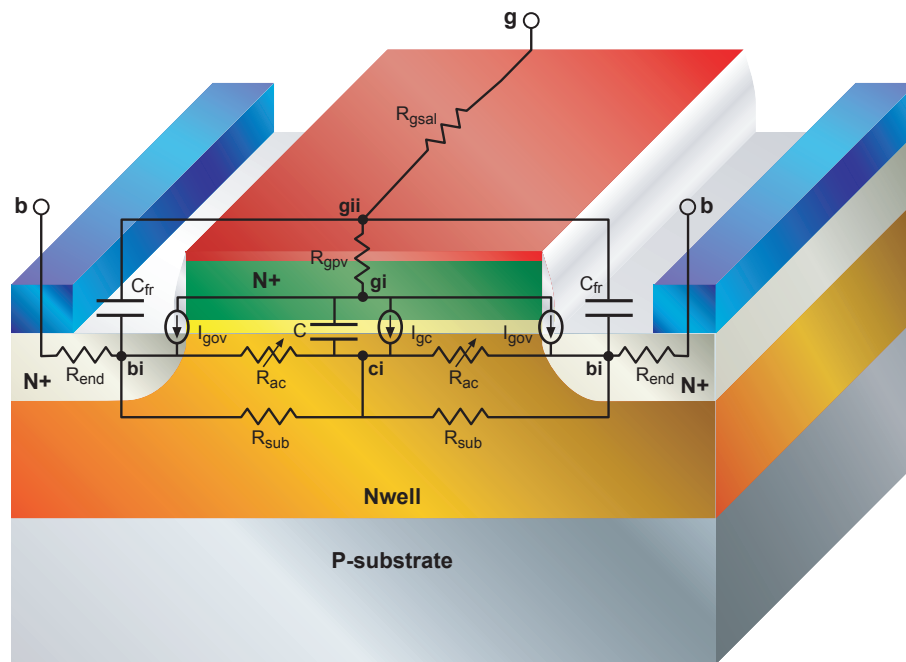


Figure 1.1: Cross section of MOS varactor with its equivalent circuit model.

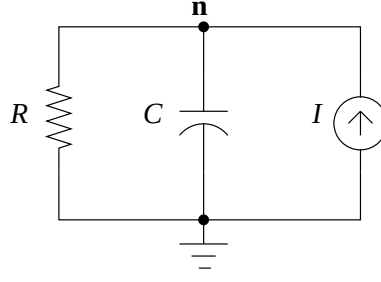


Figure 1.2: An RC circuit model used in the varactor code.

1.2 An RC Circuit Model for Inversion Charge

Figure 1.2 shows an equivalent circuit representation for how the time dependence of the inversion charge q_i , which is the voltage on the node **n**, is formulated in Verilog-A. This implements the relaxation time approximation (RTA) detailed in [4,5]

$$q_i = q_{i0} - \mathbf{TAU} \cdot \frac{dq_i}{dt} \quad (1.1)$$

The current source value is q_{i0} , the quasi-static inversion charge, $R = 1\Omega$, and $C = \mathbf{TAU} \text{ F}$.

1.3 Differences from the PSP MOSFET Model

Although the varactor model has inherited the surface potential algorithm from the PSP model [1], there are differences between them.

1. There is no position dependence of the surface potential in the varactor.
2. Dynamic formation of the inversion region is included in the varactor model [4,5].
3. Different (more elaborate) parasitics are included in the varactor model.
4. The varactor model allows different combinations of gate and well polarities.
5. More elaborate polysilicon space charge region models are included in the varactor model (in PSP, there is no poly space charge effect in accumulation).
6. The varactor model is global; there is no local model.
7. The varactor model has a simpler parameter clamping procedure.

Section 2

Constants and Parameters

In this section, the symbolic representation, value, and description of various physical constants and parameters used in the varactor model are listed. Throughout this document, varactor parameter names are all printed in boldface capitals. Parameters for temperature scaling start with ‘ST’, followed by the name of the parameter to which the temperature scaling applies.

2.1 Circuit Simulator Variables

The definitions of the external electrical variables and the simulation temperature are shown in the following table.

Symbol	Unit	Description
V_C	V	Potential across capacitor C or between nodes gi and ci
$V_{b,ov}$	V	Potential between nodes gi and bi
T_A	°C	Ambient circuit temperature

2.2 Model Constants

Physical and numerical constants used in the model are listed in the following table.

No.	Symbol	Unit	Value	Description
1	k_B	J/K	$1.3806505 \cdot 10^{-23}$	Boltzmann constant
2	$\hbar$	J s	$1.05457168 \cdot 10^{-34}$	Reduced Planck constant
3	q	C	$1.6021918 \cdot 10^{-19}$	Elementary unit charge
4	m_0	kg	$9.1093826 \cdot 10^{-31}$	Electron rest mass
5	ϵ_{ox}	F/m	$3.453 \cdot 10^{-11}$	Absolute permittivity of oxide
6	ϵ_{si}	F/m	$1.045 \cdot 10^{-10}$	Absolute permittivity of silicon

continued on next page...

... continued from previous page

No.	Symbol	Unit	Value	Description
7	QM_N	$V m^{\frac{4}{3}} C^{-\frac{2}{3}}$	5.951993	Constant of quantum-mechanical behavior of electrons
8	QM_P	$V m^{\frac{4}{3}} C^{-\frac{2}{3}}$	7.448711	Constant of quantum-mechanical behavior of holes
9	k_{se1}		$2.3025850929940458 \cdot 10^2$	Constant used in safe exponential function expl
10	k_{se2}		$4.6051701859880916 \cdot 10^2$	Constant used for exponentials

2.3 Instance Parameters

No.	Name	Unit	Default	Min.	Max.	Description
1	L	m	10^{-6}	0	–	Design length of varactor
2	W	m	10^{-6}	0	–	Design width of varactor
3	m		1	0	–	Multiplicity factor
4	NGCON		1	1	2	Number of gate contacts
5	DTA (DTEMP)	°C	0	–	–	Local temperature offset with respect to ambient circuit temperature

2.4 Special Model Parameters

No.	Name	Unit	Default	Min.	Max.	Description
1	Version		1.3	N/A	N/A	Model version
2	Revision		1	N/A	N/A	Model revision
3	TMIN	°C	-100	-250	21	Minimum ambient temperature
4	TMAX	°C	500	21	1000	Maximum ambient temperature
5	VMAX	V	10000	0	–	Maximum voltage across node g and b

2.5 Model Parameters

No.	Name	Unit	Default	Min.	Max.	Description
1	LEVEL		1000	N/A	N/A	Model level
2	TR (TREF)	°C	21	−250	1000	Nominal (reference) temperature
3	LMIN	m	10^{-8}	0	—	Minimum allowed drawn length
4	LMAX	m	$9.9 \cdot 10^{-8}$	0	—	Maximum allowed drawn length
5	WMIN	m	10^{-8}	0	—	Minimum allowed drawn width
6	WMAX	m	$9.9 \cdot 10^{-8}$	0	—	Maximum allowed drawn width
7	TOXO	m	$2 \cdot 10^{-9}$	$5 \cdot 10^{-10}$	$2 \cdot 10^{-6}$	Oxide thickness
8	EPSROXO		3.9	1.0	—	Relative permittivity
9	VFBO	V	0.0	—	—	Flat-band voltage ¹
10	NSUBO	m ^{−3}	$3 \cdot 10^{23}$	10^{18}	10^{25}	Substrate doping level
11	MNSUBO		1	1	10	Maximum change in absolute doping, limited to 1 order of magnitude increase
12	DNSUBO		0	0	100	Doping profile slope parameter
13	VNSUBO		0	−5	5	Doping profile corner voltage parameter
14	NSLPO		0.1	0.1	1	Doping profile smoothing parameter
15	DLQ	m	0	—	—	Length delta for capacitor size
16	DWQ	m	0	—	—	Width delta for capacitor size
17	DWR	m	0	—	—	Width delta for substrate resistance calculation
18	CFRL	F/m	0	0	—	Fringing capacitance in length direction
19	CFRW	F/m	0	0	—	Fringing capacitance in width direction
20	RSHG	Ω/sq	1	0	—	Gate sheet resistance
21	RPV	Ω·m ²	0	0	—	Vertical resistance down through gate
22	REND	Ω·m	10^{-4}	0	—	End resistance (extrinsic well res. plus vertical contact res. to well) per width

continued on next page...

¹In PSP, **VFBO** corresponds to NMOS and is negative of the actual value for PMOS. For varactor, one specifies the actual value.

... continued from previous page

No.	Name	Unit	Default	Min.	Max.	Description
23	RSHS	Ω/sq	1000	0	10000	Substrate sheet resistance
24	UAC	$\text{m}^2/\text{V}/\text{s}$	$5 \cdot 10^{-2}$	0	—	Accumulation layer zero-bias mobility
25	UACRED	V^{-1}	0	0	—	accumulation layer mobility degradation factor
26	STVFB	V/K	0	—	—	Temperature dependence of V_{fb}
27	STRSHG		0	—	—	Temperature dependence of R_{shg}
28	STRPV		0	—	—	Temperature dependence of R_{pv}
29	STREND		0	—	—	Temperature dependence of R_{end}
30	STRSHS		0	—	—	Temperature dependence of R_{shs}
31	STUAC		0	—	—	Temperature dependence of U_{ac}
32	FETA		1.0	0	—	Effective field parameter
Switch or switch-like parameters						
33	SWRES		1	0	1	Switch to control series resistance: 0→ exclude; 1→ include
34	TYPE		−1	−1	1	Substrate doping type: -1→ n-type; +1→ p-type
35	TYPEP		−1	−1	1	Polysilicon doping type: -1→ n-type; +1→ p-type
36	TAU	s	0.1	0	10	Time constant for inversion charge recombination/generation
37	NPO	m^{-3}	10^{27}	10^{24}	10^{27}	Polysilicon doping level
38	QMC		1	0	—	Quantum mechanical correction factor
39	SWIGATE		0	0	1	Flag for gate current: 0→turn off; 1→turn on
40	SWQINV		1	0	1	Flag for inversion charge: 0→turn off; 1→turn on
41	RACNOISE		1	0	2	Rac noise model selector: 0→turn off; 1→use bias-independent conductance; 2→use bias-dependent conductance

continued on next page...

... continued from previous page

No.	Name	Unit	Default	Min.	Max.	Description
Additional model parameters for gate tunneling currents						
42	CHIBO	V	3.1	1.0	—	Tunneling barrier height for electrons
43	CHIBPO	V	4.5	1.0	—	Tunneling barrier height for holes
44	LOV	m	0	0	—	Overlap length
45	NOVO	m^{-3}	$5 \cdot 10^{25}$	10^{22}	10^{26}	Effective doping level of overlap regions
46	IGINVLW	A	0	0	—	ECB gate channel current pre-factor for $1 \mu\text{m}^2$ channel area
47	IGOVW	A	0	0	—	ECB gate overlap current pre-factor for $1 \mu\text{m}$ wide gate overlap region
48	GCOO		0	−10	10	ECB gate tunneling energy adjustment
49	GC2O		0.375	0	10	ECB gate current slope factor
50	GC3O		0.063	−10	10	ECB gate current curvature factor
51	IGCHVLW	A	0	0	—	HVB gate channel current pre-factor for $1 \mu\text{m}^2$ channel area
52	IGOVHVV	A	0	0	—	HVB gate overlap current pre-factor for $1 \mu\text{m}$ wide gate overlap region
53	GCOHVO		0	−10	10	HVB gate tunneling energy adjustment
54	GC2HVO		0.375	0	10	HVB gate current slope factor
55	GC3HVO		0.063	−10	10	HVB gate current curvature factor
56	IGMAX	A	10^{-5}	0	—	Maximum gate current

Section 3

Parameter Initialization

In this section, internal parameters, such as effective channel length and width, oxide capacitance, and body factors, are initialized or calculated to prepare for the calculation of the surface potential, charges, and currents in the next section.

3.1 General Parameters

$$L_{\text{eff}} = L + \mathbf{DLQ} \quad (3.1)$$

$$W_{\text{eff}} = W + \mathbf{DWQ} \quad (3.2)$$

3.2 Oxide Capacitance and Body Factors

$$C_{\text{ox}} = \epsilon_{\text{ox}} \cdot (\mathbf{EPSROXO}/3.9)/\mathbf{TOXO} \quad (3.3)$$

$$\gamma_{\text{s}} = \sqrt{2 \cdot q \cdot \epsilon_{\text{si}} \cdot \mathbf{NSUBO}}/C_{\text{ox}} \quad (3.4)$$

$$\gamma_{\text{p}} = \sqrt{2 \cdot q \cdot \epsilon_{\text{si}} \cdot \mathbf{NPO}}/C_{\text{ox}} \quad (3.5)$$

$$\gamma_{\text{ov, s}} = \sqrt{2 \cdot q \cdot \epsilon_{\text{si}} \cdot \mathbf{NOVO}}/C_{\text{ox}} \quad (3.6)$$

$$\text{If } \mathbf{QMC} > 0 \quad q_{\text{q}} = \begin{cases} 0.4 \cdot QM_{\text{N}} \cdot \mathbf{QMC} \cdot (C_{\text{ox}})^{2/3}, & \text{if } \mathbf{TYPE} > 0 \\ 0.4 \cdot QM_{\text{P}} \cdot \mathbf{QMC} \cdot (C_{\text{ox}})^{2/3}, & \text{otherwise} \end{cases} \quad (3.7)$$

$$\text{else } q_{\text{q}} = 0 \quad (3.8)$$

$$\eta_{\mu} = \begin{cases} 0.5 \cdot \mathbf{FETA}, & \text{if } \mathbf{TYPE} > 0 \\ \frac{1}{3} \cdot \mathbf{FETA}, & \text{otherwise} \end{cases} \quad (3.9)$$

$$\text{norm}_{\text{tox}} = \mathbf{TOXO}/10^{-9}; \quad (3.10)$$

3.3 Temperature-Related Parameters

$$\text{TR1} = \begin{cases} \text{TR}, & \text{if } \text{TR} \geq -273 \\ -273, & \text{if } \text{TR} < -273 \end{cases} \quad (3.11)$$

$$T_{\text{KR}} = 273.15 + \text{TR1} \quad (3.12)$$

$$T_{\text{KD}} = T_{\text{A}} + \text{DTA} \quad (3.13)$$

$$\Delta T = T_{\text{KD}} - T_{\text{KR}} \quad (3.14)$$

$$\phi_{\text{T}} = k_{\text{B}} \cdot T_{\text{KD}} / q \quad (3.15)$$

$$q_{\text{lim2}} = 100 \cdot \phi_{\text{T}}^2 \quad (3.16)$$

$$E_{\text{g}} = 1.179 - T_{\text{KD}} \cdot (9.025 \cdot 10^{-5} + 3.05 \cdot 10^{-7} \cdot T_{\text{KD}}) \quad (3.17)$$

$$r_{\text{T}} = (1.045 + 4.5 \cdot 10^{-4} \cdot T_{\text{KD}}) \cdot (0.523 + 1.4 \cdot 10^{-3} \cdot T_{\text{KD}} - 1.48 \cdot 10^{-6} \cdot T_{\text{KD}}^2) \cdot \frac{T_{\text{KD}}^2}{90000} \quad (3.18)$$

$$\text{INV}_{\text{ni}} = 4 \cdot 10^{-26} \cdot r_{\text{T}}^{-0.75} \quad (3.19)$$

$$\phi_{\text{b}} = E_{\text{g}} + 2 \cdot \phi_{\text{T}} \cdot \ln(\text{NSUBO} \cdot \text{INV}_{\text{ni}}) \quad (3.20)$$

$$V_{\text{fb, T}} = \text{VFBO} + \Delta T \cdot \text{STVFB} \quad (3.21)$$

$$R_{\text{shg, T}} = \text{RSHG} \cdot \left(\frac{T_{\text{KR}}}{T_{\text{KD}}} \right)^{\text{STRSHG}} \quad (3.22)$$

$$R_{\text{pv, T}} = \text{RPV} \cdot \left(\frac{T_{\text{KR}}}{T_{\text{KD}}} \right)^{\text{STRPV}} \quad (3.23)$$

$$R_{\text{end, T}} = \text{REND} \cdot \left(\frac{T_{\text{KR}}}{T_{\text{KD}}} \right)^{\text{STREND}} \quad (3.24)$$

$$R_{\text{shs, T}} = \text{RSHS} \cdot \left(\frac{T_{\text{KR}}}{T_{\text{KD}}} \right)^{\text{STRSHS}} \quad (3.25)$$

$$U_{\text{ac, T}} = \text{UAC} \cdot \left(\frac{T_{\text{KD}}}{T_{\text{KR}}} \right)^{\text{STUAC}} \quad (3.26)$$

3.4 Parameters Related to Polysilicon and Overlap Regions

$$G_{\text{p}} = \gamma_{\text{p}} / \sqrt{\phi_{\text{T}}} \quad (3.27)$$

$$\xi_{\text{p}} = 1 + G_{\text{p}} / \sqrt{2} \quad (3.28)$$

$$x_{\text{mrp}} = 10^{-5} \cdot \xi_{\text{p}} \quad (3.29)$$

$$\phi_{\text{p}} = E_{\text{g}} + 2 \cdot \phi_{\text{T}} \cdot \ln(\mathbf{NPO} \cdot \text{INV}_{\text{ni}}) \quad (3.30)$$

$$x_{\text{np}} = \phi_{\text{p}} / \phi_{\text{T}} \quad (3.31)$$

$$\Delta_{\text{np}} = \begin{cases} \exp(-x_{\text{np}}), & \text{if } x_{\text{np}} < k_{\text{se2}} \\ \frac{10^{-200}}{\text{P3}(x_{\text{np}} - k_{\text{se2}})}, & \text{otherwise} \end{cases} \quad (3.32)$$

$$G_{\text{ov},s} = \gamma_{\text{ov},s} / \sqrt{\phi_{\text{T}}} \quad (3.33)$$

$$\xi_{\text{ov},s} = 1 + G_{\text{ov},s} / \sqrt{2} \quad (3.34)$$

$$x_{\text{mrgov},s} = 10^{-5} \cdot \xi_{\text{ov},s} \quad (3.35)$$

$$\phi_{\text{b,ov}} = E_{\text{g}} + 6 \cdot \phi_{\text{T}} \quad (3.36)$$

$$x_1 = 1.25 \quad (3.37)$$

$$x_{\text{g1,ov}} = x_1 + G_{\text{ov},s} \cdot \sqrt{\exp(-x_1) + x_1 - 1} \quad (3.38)$$

3.5 Fringing Capacitance

$$C_{\text{fr}} = 2 \cdot (\mathbf{CFRW} \cdot W + \mathbf{CFRL} \cdot L) \quad (3.39)$$

3.6 Resistances

$$\begin{aligned}
 \text{If } \mathbf{SWRES} = \text{true} \left\{ \begin{aligned}
 R_{\text{gsal}} &= \frac{R_{\text{shg}, \text{T}} \cdot W}{L \cdot [3 + 9 \cdot (\mathbf{NGCON} - 1)]} \\
 R_{\text{gpv}} &= \frac{R_{\text{pv}, \text{T}}}{W \cdot L} \\
 R_{\text{end}} &= \frac{R_{\text{end}, \text{T}}}{2 \cdot (W + \mathbf{DWR})} \\
 R_{\text{sub}} &= \frac{R_{\text{shs}, \text{T}} \cdot L}{12 \cdot (W + \mathbf{DWR})} \\
 R_{\text{gsal}} &= \text{'CLIP_BOTH'}(R_{\text{gsal}}, 1.0e - 03, 1e01) \\
 R_{\text{gpv}} &= \text{'CLIP_BOTH'}(R_{\text{gpv}}, 1.0e - 03, 1e02) \\
 R_{\text{end}} &= \text{'CLIP_BOTH'}(R_{\text{end}}, 1.0e - 03, 1e01) \\
 R_{\text{sub}} &= \text{'CLIP_BOTH'}(R_{\text{sub}}, 1.0e - 03, 1e03) \\
 U_{\text{ac}, \text{T}} &= \text{'CLIP_BOTH'}(U_{\text{ac}, \text{T}}, 1.0e - 03, 2e01) \\
 G_{\text{gsal}} &= 1/R_{\text{gsal}} \\
 G_{\text{gpv}} &= 1/R_{\text{gpv}} \\
 G_{\text{end}} &= 1/R_{\text{end}} \\
 G_{\text{sub}} &= 1/R_{\text{sub}} \\
 G_{\text{ac0}} &= 12 \cdot U_{\text{ac}, \text{T}} \cdot W/L
 \end{aligned} \right. \tag{3.40}
 \end{aligned}$$

$$\begin{aligned}
 \text{If } \mathbf{SWRES} = \text{false} \left\{ \begin{aligned}
 G_{\text{gsal}} &= 0.0 \\
 G_{\text{gpv}} &= 0.0 \\
 G_{\text{end}} &= 0.0 \\
 G_{\text{sub}} &= 0.0 \\
 G_{\text{ac0}} &= 0.0
 \end{aligned} \right. \tag{3.41}
 \end{aligned}$$

3.7 Noise Parameter

$$nt0 = 4 \cdot k_{\text{B}} \cdot T_{\text{KD}} \tag{3.42}$$

3.8 Gate Tunneling Parameters

$$\begin{aligned}
 \text{If SWIGATE} = \text{true} \left\{ \begin{aligned}
 I_{\text{ginv}} &= \mathbf{IGINVLW} \cdot W_{\text{eff}} \cdot L_{\text{eff}} \cdot 10^{12} \cdot \left(\frac{T_{\text{KD}}}{T_{\text{KR}}} \right)^{\text{STIG}} \\
 I_{\text{gov}} &= 2 \cdot \mathbf{IGOVW} \cdot \mathbf{LOV} \cdot W_{\text{eff}} \cdot 10^{12} \cdot \left(\frac{T_{\text{KD}}}{T_{\text{KR}}} \right)^{\text{STIG}} \\
 I_{\text{gcHVB}} &= \mathbf{IGCHVLW} \cdot W_{\text{eff}} \cdot L_{\text{eff}} \cdot 10^{12} \cdot \left(\frac{T_{\text{KD}}}{T_{\text{KR}}} \right)^{\text{STIG}} \\
 I_{\text{govHVB}} &= 2 \cdot \mathbf{IGOVHVV} \cdot \mathbf{LOV} \cdot W_{\text{eff}} \cdot 10^{12} \cdot \left(\frac{T_{\text{KD}}}{T_{\text{KR}}} \right)^{\text{STIG}} \\
 \text{INV}_{\text{CHIB}} &= 1/\mathbf{CHIBO} \\
 \text{INV}_{\text{CHIB, HVB}} &= 1/\mathbf{CHIBPO} \\
 B_{\text{CH}} &= (4/3) \cdot \mathbf{TOXO} \cdot \sqrt{2 \cdot q \cdot m_0 \cdot \mathbf{CHIBO}} / \hbar \\
 B_{\text{OV}} &= B_{\text{CH}} \\
 B_{\text{CH, HVB}} &= (4/3) \cdot \mathbf{TOXO} \cdot \sqrt{2 \cdot q \cdot m_0 \cdot \mathbf{CHIBPO}} / \hbar \\
 B_{\text{OV, HVB}} &= B_{\text{CH, HVB}} \\
 \text{If } \mathbf{GC3O} < 0, & Q_{\text{CQ}} = -0.495 \cdot \mathbf{GC2O}/\mathbf{GC3O}, \\
 \text{else } & Q_{\text{CQ}} = 0.0, \text{ end} \\
 \text{If } \mathbf{GC3HVO} < 0, & Q_{\text{CQ, HVB}} = -0.495 \cdot \mathbf{GC2HVO}/\mathbf{GC3HVO}, \\
 \text{else } & Q_{\text{CQ, HVB}} = 0.0, \text{ end} \\
 \alpha_{\text{b, s}} &= 0.5 \cdot (E_{\text{g}} + \mathbf{TYPE} \cdot \phi_{\text{b}}) \\
 \alpha_{\text{b, ov}} &= 0.5 \cdot (E_{\text{g}} + \mathbf{TYPE} \cdot \phi_{\text{b, ov}}) \\
 D_{\text{ch}} &= \mathbf{GCOO} \cdot \phi_{\text{T}} \\
 D_{\text{ch, HVB}} &= \mathbf{GCOHVO} \cdot \phi_{\text{T}}
 \end{aligned} \right. \tag{3.43}
 \end{aligned}$$

$$\begin{aligned}
 \text{If SWIGATE} = \text{false} \left\{ \begin{aligned}
 I_{\text{ginv}} &= 0 \\
 I_{\text{gov}} &= 0 \\
 I_{\text{gcHVB}} &= 0 \\
 I_{\text{govHVB}} &= 0 \\
 \text{INV}_{\text{CHIB}} &= 0.1 \\
 \text{INV}_{\text{CHIB, HVB}} &= 0.1 \\
 B_{\text{CH}} &= B_{\text{OV}} = 0 \\
 B_{\text{CH, HVB}} &= B_{\text{OV, HVB}} = 0 \\
 Q_{\text{CQ}} &= 0 \\
 Q_{\text{CQ, HVB}} &= 0 \\
 \alpha_{\text{b, s}} &= 0 \\
 \alpha_{\text{b, ov}} &= 0 \\
 D_{\text{ch}} &= 0 \\
 D_{\text{ch, HVB}} &= 0
 \end{aligned} \right. \tag{3.44}
 \end{aligned}$$

Section 4

Model Equations

4.1 Static Evaluations

$$N_{b, \text{lim}} = \text{NSUBO} \cdot \text{MNSUBO} \quad (4.1)$$

$$N_{b1} = \text{NSUBO} \cdot [1 + \text{DNSUBO} \cdot \text{MAXA}(\text{TYPE} \cdot (V_C - \text{VNSUBO}), 0, \text{NSLPO})], \quad (4.2)$$

where V_C is the voltage across the capacitor C in in Figure 1.1.

$$N_{b, v} = \text{NSUBO} \cdot \text{MINA}(N_{b1}/\text{NSUBO}, \text{MNSUBO}, 10^{-6}) \quad (4.3)$$

$$\text{norm}_{\text{sub}} = N_{b, v}/10^{23} \quad (4.4)$$

$$\phi_{b1} = E_g + 2 \cdot \phi_T \cdot \ln(N_{b, v} \cdot \text{INV}_{\text{ni}}) \quad (4.5)$$

$$\gamma_{s1} = \sqrt{2 \cdot q \cdot \epsilon_{\text{si}} \cdot N_{b, v} / C_{\text{ox}}} \quad (4.6)$$

$$\text{If } \mathbf{QMC} > 0 \left\{ \begin{array}{l} q_{b0} = \sqrt{\gamma_s^2 \cdot \phi_b} \\ d_{\phi b q} = 0.75 \cdot q_q \cdot (q_{b0})^{2/3} \\ \phi_b = \phi_{b1} + d_{\phi b q} \\ \gamma_s = \gamma_{s1} \cdot \left(1 + \frac{4}{3} \frac{d_{\phi b q}}{q_{b0}}\right) \end{array} \right. \quad (4.7)$$

$$G_s = \gamma_s / \sqrt{\phi_T} \quad (4.8)$$

$$\xi_s = 1 + G_s / \sqrt{2} \quad (4.9)$$

$$x_{\text{mrsg}} = 10^{-5} \cdot \xi_s \quad (4.10)$$

$$x_{\text{ns}} = \phi_b / \phi_T \quad (4.11)$$

$$\Delta_{\text{ns}} = \begin{cases} \exp(-x_{\text{ns}}), & \text{if } x_{\text{ns}} < k_{\text{se2}} \\ \frac{10^{-200}}{\text{P3}(x_{\text{ns}} - k_{\text{se2}})}, & \text{otherwise} \end{cases} \quad (4.12)$$

$$x_{\text{g1}} = x_1 + G_s \cdot \sqrt{\exp(-x_1) + x_1 - 1} \quad (4.13)$$

$$x_{\text{g1, ov}} = x_1 + G_{\text{ov, s}} \cdot \sqrt{\exp(-x_1) + x_1 - 1} \quad (4.14)$$

4.2 Calculation of the Surface Potential at the Channel Side

A macro Φ_s is defined to calculate the surface potential at the channel side with output x_s and inputs: x_g , x_{ns} , Δ_{ns} , G , G^2 , G^{-2} , ξ , ξ^{-1} , x_{mrg} . The definitions of the auxiliary functions MINA, MAXA, σ_1 , σ_2 , and others can be found in Appendix A.

$$\text{if } x_g < -x_{\text{mrg}} \left\{ \begin{array}{l} y_g = -x_g \\ z = 1.25 \cdot y_g / \xi \\ \eta = \left[z + 10 - \sqrt{(z - 6)^2 + 64} \right] / 2 \\ a = (y_g - \eta)^2 + G^2 \cdot (\eta + 1) \\ c = 2 \cdot (y_g - \eta) - G^2 \\ \tau = -\eta + \ln(a / G^2) \\ y_0 = \sigma_1(a, c, \tau, \eta) \\ \Delta_0 = \text{expl}_{\text{high}}(y_0) \\ p = 2 \cdot (y_g - y_0) + G^2 \cdot [\Delta_0 - 1 + \Delta_{\text{ns}} \cdot (1 - 1/\Delta_0)] \\ q = (y_g - y_0)^2 + G^2 \cdot [y_0 - \Delta_0 + 1 + \Delta_{\text{ns}} \cdot (1 - 1/\Delta_0 - 2 \cdot y_0)] \\ x_s = -y_0 - \frac{2 \cdot q}{p + \sqrt{p^2 - 2 \cdot q \cdot [2 - G^2 \cdot (\Delta_0 + \Delta_{\text{ns}}/\Delta_0)]}} \end{array} \right. \quad (4.15)$$

$$\text{if } |x_g| \leq x_{\text{mrg}} \left\{ x_s = \frac{x_g}{\xi} \cdot \left[1 + G \cdot x_g \cdot \frac{1 - \Delta_{\text{ns}}}{\xi^2 \cdot 6 \cdot \sqrt{2}} \right] \right. \quad (4.16)$$

$$\text{if } x_g > x_{\text{mrg}} \left\{ \begin{array}{l}
\hat{x}_{g1} = x_1 + G \cdot \sqrt{\exp(-x_1) + x_1 - 1} \\
\bar{x} = \frac{x_g}{\xi} \cdot [1 + x_g \cdot (\xi \cdot x_1 - \hat{x}_{g1}) / \hat{x}_{g1}^2] \\
x_0 = x_g + G^2/2 - G \cdot \sqrt{x_g + G^2/4 - 1 + \text{expl}_{\text{low}}(-\bar{x})} \\
b_x = x_{\text{ns}} + 3 \\
\eta = \text{MINA}(x_0, b_x, 5) - (b_x - \sqrt{b_x^2 + 5}) / 2 \\
a = (x_g - \eta)^2 - G^2 \cdot [\exp(-\eta) + \eta - 1 - \Delta_{\text{ns}} \cdot (\eta + 1)] \\
b = 1 - G^2/2 \cdot \exp(-\eta) \\
c = 2 \cdot (x_g - \eta) + G^2 \cdot [1 - \exp(-\eta) - \Delta_{\text{ns}}] \\
\tau = x_{\text{ns}} - \eta + \ln(a/G^2) \\
y_0 = \sigma_2(a, b, c, \tau, \eta) \\
\Delta_0 = \begin{cases} \exp(y_0), & \text{if } y_0 < k_{\text{se1}} \\ \exp(y_0 - x_{\text{ns}}), & \text{if } y_0 > x_{\text{ns}} - k_{\text{se1}} \\ \frac{10^{-100}}{\text{P3}(x_{\text{ns}} - y_0 - k_{\text{se1}})}, & \text{otherwise} \end{cases} \\
p = 2 \cdot (x_g - y_0) + G^2 \cdot [1 - 1/\Delta_0 + \Delta_{\text{ns}} \cdot (\Delta_0 - 1)] \\
q = (x_g - y_0)^2 - G^2 \cdot [y_0 + 1/\Delta_0 - 1 + \Delta_{\text{ns}} \cdot (\Delta_0 - y_0 - 1)] \\
x_s = y_0 + \frac{2 \cdot q}{p + \sqrt{p^2 - 2 \cdot q \cdot [2 - G^2 \cdot (1/\Delta_0 + \Delta_{\text{ns}} \cdot \Delta_0)]}}
\end{array} \right. \quad (4.17)$$

4.3 Calculation of the Surface Potential in the Overlap Regions

A macro Φ_{ov} is defined to calculate the surface potential in the overlap regions with output x_{ov} and inputs:

$x_{\text{g}}, G_{\text{ov}}, G_{\text{ov}}^2, x_{\text{mrgov}}, \xi_{\text{ov}}, x_{\text{g1}}.$

$$x_{\text{ov}}(x_{\text{g}}) = \begin{cases} \text{if } x_{\text{g}} < -x_{\text{mrgov}} \left\{ \begin{array}{l} y_{\text{g}} = -x_{\text{g}} \\ z = x_1 \cdot y_{\text{g}} / \xi_{\text{ov}} \\ \eta = \left[z + 10 - \sqrt{(z - 6)^2 + 64} \right] / 2 \\ a = (y_{\text{g}} - \eta)^2 + G_{\text{ov}}^2 \cdot (\eta + 1) \\ c = 2 \cdot (y_{\text{g}} - \eta) - G_{\text{ov}}^2 \\ \tau = -\eta + \ln(a / G_{\text{ov}}^2) \\ y_0 = \sigma_1(a, c, \tau, \eta) \\ \Delta_0 = \exp(y_0) \\ p = 2 \cdot (y_{\text{g}} - y_0) + G_{\text{ov}}^2 \cdot (\Delta_0 - 1) \\ q = (y_{\text{g}} - y_0)^2 + G_{\text{ov}}^2 \cdot (y_0 - \Delta_0 + 1) \\ x_{\text{ov}} = -y_0 - \frac{2 \cdot q}{p + \sqrt{p^2 - 2 \cdot q \cdot (2 - G_{\text{ov}}^2 \cdot \Delta_0)}} \end{array} \right. \\ \\ \text{if } |x_{\text{g}}| < x_{\text{mrgov}} \left\{ \begin{array}{l} x_{\text{ov}} = x_{\text{g}} / \xi_{\text{ov}} \end{array} \right. \\ \\ \text{if } x_{\text{g}} > x_{\text{mrgov}} \left\{ \begin{array}{l} \bar{x} = x_{\text{g}} / \xi_{\text{ov}} \cdot \left[1 + x_{\text{g}} \cdot (\xi_{\text{ov}} \cdot x_1 - x_{\text{g1}}) / x_{\text{g1}}^2 \right] \\ \omega = \begin{cases} 1 - \exp(-\bar{x}), & \text{if } \bar{x} < k_{\text{se2}} \\ 1 - \frac{10^{-200}}{\text{P3}(\bar{x} - k_{\text{se2}})}, & \text{otherwise} \end{cases} \\ x_0 = x_{\text{g}} + G_{\text{ov}}^2 / 2 - G_{\text{ov}} \cdot \sqrt{x_{\text{g}} + G_{\text{ov}}^2 / 4 - \omega} \\ \Delta_0 = \begin{cases} \exp(-x_0), & \text{if } x_0 < k_{\text{se2}} \\ \frac{10^{-200}}{\text{P3}(x_0 - k_{\text{se2}})}, & \text{otherwise} \end{cases} \\ p = 2 \cdot (x_{\text{g}} - x_0) + G_{\text{ov}}^2 \cdot (1 - \Delta_0) \\ q = (x_{\text{g}} - x_0)^2 - G_{\text{ov}}^2 \cdot (x_0 + \Delta_0 - 1) \\ x_{\text{ov}} = x_0 + \frac{2 \cdot q}{p + \sqrt{p^2 - 2 \cdot q \cdot (2 - G_{\text{ov}}^2 \cdot \Delta_0)}} \end{array} \right. \end{cases} \quad (4.18)$$

4.4 Surface Potential without Poly Effect

$$V_{\text{gb1}} = \text{TYPE} \cdot (V_{\text{C}} - V_{\text{fb,T}}) \quad (4.19)$$

$$x_{\text{g}} = V_{\text{gb1}} / \phi_{\text{T}} \quad (4.20)$$

The macro Φ_{s} (defined in Section 4.2) calculates the normalized surface potential (i.e., $\phi_{\text{s}} / \phi_{\text{T}}$).

$$x_{\text{s0}} = \Phi_{\text{s}}(x_{\text{g}}, x_{\text{ns}}, \Delta_{\text{ns}}, G_{\text{s}}, G_{\text{s}}^2, G_{\text{s}}^{-2}, \xi_{\text{s}}, \xi_{\text{s}}^{-1}, x_{\text{mrgrs}}) \quad (4.21)$$

$$\psi_{\text{s0}} = x_{\text{s0}} \cdot \phi_{\text{T}}. \quad (4.22)$$

4.5 Surface Potential with Poly Effect

The polysilicon space charge region is one area where the varactor model differs significantly from PSP model. In the varactor model, polysilicon is allowed to enter inversion region. Furthermore, the effect of finite doping of the polysilicon is included for all gate biases (in PSP the polysilicon effect is not included when the device operates in accumulation).

Note also that even when the frequency-dependent inversion layer formation is activated in the varactor model of the active region, the polysilicon surface potential is still computed using the quasi-static approximation.

When $\text{NPO} \geq 10^{27}$, $\psi_{\text{p0}} = 0$. The following equations calculate the poly surface potential when $\text{NPO} < 10^{27}$.

$$x_{\text{gp}} = -\text{TYPE} \cdot \text{TYPEP} \cdot (V_{\text{gb1}} - \psi_{\text{s0}}) / \phi_{\text{T}} \quad (4.23)$$

$$x_{\text{p0}} = \Phi_{\text{s}}(x_{\text{gp}}, x_{\text{np}}, \Delta_{\text{np}}, G_{\text{p}}, G_{\text{p}}^2, G_{\text{p}}^{-2}, \xi_{\text{p}}, \xi_{\text{p}}^{-1}, x_{\text{mrgrp}}) \quad (4.24)$$

$$\psi_{\text{p0}} = -\text{TYPE} \cdot \text{TYPEP} \cdot x_{\text{p0}} \cdot \phi_{\text{T}} \quad (4.25)$$

$$x_{\text{g}} = (V_{\text{gb1}} - \psi_{\text{p0}}) / \phi_{\text{T}} \quad (4.26)$$

$$x_{\text{s0}} = \Phi_{\text{s}}(x_{\text{g}}, x_{\text{ns}}, \Delta_{\text{ns}}, G_{\text{s}}, G_{\text{s}}^2, G_{\text{s}}^{-2}, \xi_{\text{s}}, \xi_{\text{s}}^{-1}, x_{\text{mrgrs}}) \quad (4.27)$$

$$\psi_{\text{s0}} = x_{\text{s0}} \cdot \phi_{\text{T}}. \quad (4.28)$$

4.6 Static Inversion Charge Calculations

The following computes surface potential-related variables and inversion charge.

$$\text{If } x_{\text{g}} \leq 0 \quad q_{\text{is}} = 0 \quad (4.29)$$

When $x_{\text{g}} > 0$ (namely the depletion and inversion regions), the following is performed to find q_{is} and q_{eff} .

$$\Delta_{\text{ls}} = 0 \quad (4.30)$$

$$\left\{ \begin{array}{l} \text{If } x_{s0} < k_{se1} \\ \text{else if } x_{s0} > x_{ns} - k_{se1} \\ \text{else} \end{array} \right\} \left\{ \begin{array}{l} \Delta_{ls1} = \exp(x_{s0}) \\ E_s = \Delta_{ls1}^{-1} \\ \Delta_{ls} = \Delta_{ns} \cdot \Delta_{ls1} \\ D_s = \Delta_{ns} \cdot (E_s^{-1} - x_{s0} - 1) \\ \Delta_{ls} = \exp(x_{s0} - x_{ns}) \\ E_s = \Delta_{ns} / \Delta_{ls} \\ D_s = \Delta_{ls} - \Delta_{ns} \cdot (x_{s0} + 1) \\ \Delta_{ls} = \frac{10^{-100}}{\text{P3}(x_{ns} - x_{s0} - k_{se1})} \\ E_s = \frac{10^{-100}}{\text{P3}(x_{s0} - k_{se1})} \\ D_s = \Delta_{ls} - \Delta_{ns} \cdot (x_{s0} + 1) \end{array} \right. \quad (4.31)$$

$$\left\{ \begin{array}{l} \text{If } x_{s0} < 10^{-5} \\ \text{else} \end{array} \right\} \left\{ \begin{array}{l} P_s = 0.5 \cdot x_{s0}^2 \cdot \left[1 - \frac{1}{3} x_{s0} \cdot (1 - 0.25 \cdot x_{s0}) \right] \\ D_s = \frac{1}{6} \Delta_{ns} \cdot x_{s0}^3 \cdot (1 + 1.75 \cdot x_{s0}) \\ S_{qs} = x_{s0} \cdot \sqrt{0.5 - \frac{1}{6} x_{s0} \cdot (1 - 0.25 \cdot x_{s0})} \\ P_s = x_{s0} + E_s - 1 \\ S_{qs} = \sqrt{P_s} \end{array} \right. \quad (4.32)$$

$$\begin{aligned} x_{gs} &= G_s \cdot \sqrt{P_s + D_s} \\ q_{is} &= \phi_{\mathbf{T}} \cdot G_s^2 \cdot D_s / (x_{gs} + G_s \cdot S_{qs}) \end{aligned} \quad (4.33)$$

The normalized static inversion charge, when **SWQINV** = 1, is

$$Q_{i0} = -q_{is} \quad (4.34)$$

When **SWQINV** = 0, the inversion charge calculation is skipped, and

$$Q_{i0} = 0 \quad (4.35)$$

4.7 Time-Dependent Silicon Surface Potential Calculation without Poly Effect

$$x_{g,t} = (V_{gb1} + V_n) / \phi_{\mathbf{T}}, \quad (4.36)$$

where V_n is the voltage at the internal time-constant node **n** in the equivalent circuit [1.2](#).

$$x_s = \Phi_{ov}(x_{g,t}, G_s, G_s^2, x_{mrgs}, \xi_s, x_{g1}), \quad (4.37)$$

where Φ_{ov} calculates the normalized surface potential and is defined in Section 4.3.

$$\psi_s = x_s \cdot \phi_T \quad (4.38)$$

4.8 Time-Dependent Poly Surface Potential Calculation Correction

If $\text{NPO} \geq 10^{27}$, $\psi_p = 0$. Otherwise, the following procedure is used.

$$x_{\text{gp}, \text{t}} = -\text{TYPE} \cdot \text{TYPEP} \cdot (V_{\text{gb1}} - \psi_s) / \phi_T \quad (4.39)$$

$$x_p = \Phi_s(x_{\text{gp}, \text{t}}, x_{\text{np}}, \Delta_{\text{np}}, G_p, G_p^2, G_p^{-2}, \xi_p, \xi_p^{-1}, x_{\text{mrp}}) \quad (4.40)$$

$$\psi_p = -\text{TYPE} \cdot \text{TYPEP} \cdot x_p \cdot \phi_T \quad (4.41)$$

$$x_{\text{g}, \text{t}} = (V_{\text{gb1}} + V_n - \psi_p) / \phi_T \quad (4.42)$$

$$x_s = \Phi_{\text{ov}}(x_{\text{g}, \text{t}}, G_s, G_s^2, x_{\text{mrgs}}, \xi_s, x_{\text{gl}}) \quad (4.43)$$

$$\psi_s = x_s \cdot \phi_T \quad (4.44)$$

4.9 Quantum Mechanical Corrections

$$\Delta_{\text{ls}} = 0 \quad (4.45)$$

$$\left\{ \begin{array}{ll} \text{If } x_s < k_{\text{se1}} & \left\{ \begin{array}{l} \Delta_{\text{ls}} = \exp(x_s) \\ E_s = \Delta_{\text{ls}}^{-1} \end{array} \right. \\ \text{else if } x_s > x_{\text{ns}} - k_{\text{se1}} & \left\{ \begin{array}{l} \Delta_{\text{ls}} = \exp(x_{\text{ns}} - x_s) \\ E_s = \Delta_{\text{ns}} \cdot \Delta_{\text{ls}} \end{array} \right. \\ \text{else} & E_s = \frac{10^{-100}}{\text{P3}(x_s - k_{\text{se1}})} \end{array} \right. \quad (4.46)$$

$$\left\{ \begin{array}{ll} \text{If } x_s < -x_{\text{mrgs}} & \left\{ \begin{array}{l} P_s = E_s + x_s - 1.0 \\ S_{\text{qs}} = -\sqrt{P_s} \end{array} \right. \\ \text{else if } \text{abs}(x_s) \leq x_{\text{mrgs}} & \left\{ \begin{array}{l} P_s = 0.5 \cdot x_s^2 \cdot \left[1 - \frac{1}{3} x_s \cdot (1 - 0.25 \cdot x_s) \right] \\ S_{\text{qs}} = x_s \cdot \sqrt{0.5 - \frac{1}{6} x_s \cdot (1 - 0.25 \cdot x_s)} \end{array} \right. \\ \text{else} & \left\{ \begin{array}{l} P_s = x_s + E_s - 1 \\ S_{\text{qs}} = \sqrt{P_s} \end{array} \right. \end{array} \right. \quad (4.47)$$

$$\begin{aligned}
q_{bs} &= \phi_T \cdot S_{qs} \cdot G_s \\
\epsilon &= 1.62 \cdot [(1 + norm_{nsub}) \cdot (1 + 0.37 \cdot norm_{tox})]^2 \cdot \left(\frac{T_{KR}}{T_{KD}}\right)^{1.5} \cdot \phi_T^2; \\
q_{eff} &= \text{MAXA}(q_{bs}, -q_{bs}, \epsilon) + \eta_\mu \cdot \text{MAXA}(-V_n, V_n, \epsilon)
\end{aligned} \tag{4.48}$$

The following $C_{ox, qm}$ calculation is for all regions ($x_g \leq 0$ and $x_g > 0$).

$$C_{ox, qm} = \begin{cases} C_{ox}, & q_q = 0 \\ \frac{C_{ox}}{1 + q_q \cdot (q_{eff}^2 + q_{lim2})^{-1/6}}, & q_q > 0 \end{cases} \tag{4.49}$$

4.10 Accumulation Resistance Bias Dependence

$$f_{rac} = \begin{cases} \phi_T \cdot \exp(-10), & \text{if } x_{s0} > 10 \\ \phi_T \cdot \exp(-x_{s0}), & \text{otherwise} \end{cases} \tag{4.50}$$

$$q_{ac} = \gamma_s \cdot C_{ox, qm} \cdot \sqrt{f_{rac}} \tag{4.51}$$

$$m_{axs} = 0.5 \cdot \left(-V_{gb1} + \sqrt{V_{gb1}^2 + 0.04}\right) \tag{4.52}$$

$$G_{ac} = G_{ac0} \cdot q_{ac} / (1 + \mathbf{UACRED} \cdot m_{axs}) \tag{4.53}$$

4.11 Gate Tunneling Current Macro

A macro I_{gate} is defined as a function of I_{gin} , I_{ginHVB} , E_g , V_{ov} , D_{ch} , $D_{ch, HVB}$, INV_{CHIB} , $INV_{CHIB, HVB}$, **GC2O**, **GC3O**, **GC2HVO**, **GC3HVO**, Q_{CQ} , $Q_{CQ, HVB}$, $I_{g, type}$, x_s , $\alpha_{b, s}$, $\alpha_{b, ov}$, ϕ_T^{-1} , **TYPEP**, **TYPE**, $V_{b, ig}$, B_{OV} , and $B_{OV, HVB}$ as shown in the following with I_{gout} as output. It is used for the calculation of gate tunneling current. See Figure 4.1 for different gate tunneling mechanisms.

$$\text{If } \mathbf{TYPEP} = 1 \left\{ \begin{aligned} &\psi_t = \text{MAXA}(0, \mathbf{TYPE} \cdot V_{ov} + D_{ch, HVB}, 0.01) \\ &z_g = \sqrt{V_{ov}^2 + 10^{-6}} \cdot INV_{CHIB, HVB} \\ &\text{If } \mathbf{GC3HVO} < 0 \\ &\quad z_g = \text{MINA}(z_g, Q_{CQ, HVB}, 10^{-6}) \\ &\text{end} \\ &\Delta_{si} = \begin{cases} \exp[-\mathbf{TYPE} \cdot x_s - (E_g - \alpha_{b, ov} + \psi_t) \cdot \phi_T^{-1}], & \text{if } I_{g, type} = 0 \\ \exp[-\mathbf{TYPE} \cdot x_s - (E_g - \alpha_{b, s} + \psi_t) \cdot \phi_T^{-1}], & \text{if } I_{g, type} = 1 \end{cases} \\ &\Delta_{gate} = \Delta_{si} \cdot \exp(\mathbf{TYPE} \cdot V_{b, ig} \cdot \phi_T^{-1}) \\ &D = \exp\{B_{OV} \cdot [-1.5 + z_g \cdot (\mathbf{GC2HVO} + \mathbf{GC3HVO} \cdot z_g)]\} \\ &I_{gout, HVB} = I_{ginHVB} \cdot D \cdot \mathbf{TYPE} \cdot \ln\left(\frac{1.0 + \Delta_{gate}}{1.0 + \Delta_{si}}\right) \end{aligned} \right. \tag{4.54}$$

$$\psi_t = \text{MINA}(0, \text{TYPE} \cdot V_{\text{ov}} + D_{\text{ch}}, 0.01) \quad (4.55)$$

$$z_g = \sqrt{V_{\text{ov}}^2 + 10^{-6}} \cdot \text{INV}_{\text{CHIB}} \quad (4.56)$$

If **GC30** < 0

$$z_g = \text{MINA}(z_g, Q_{\text{CQ}}, 10^{-6}) \quad (4.57)$$

$$\Delta_{\text{si}} = \begin{cases} \exp(\text{TYPE} \cdot x_s + (-\alpha_{\text{b,ov}} + \psi_t) \cdot \phi_{\text{T}}^{-1}), & \text{if } I_{\text{g,type}} = 0 \\ \exp(\text{TYPE} \cdot x_s + (-\alpha_{\text{b,s}} + \psi_t) \cdot \phi_{\text{T}}^{-1}), & \text{if } I_{\text{g,type}} = 1 \end{cases} \quad (4.58)$$

$$\Delta_{\text{gate}} = \Delta_{\text{si}} \cdot \exp(-\text{TYPE} \cdot V_{\text{b,ig}} \cdot \phi_{\text{T}}^{-1}) \quad (4.59)$$

$$D = \exp\{B_{\text{OV}} \cdot [-1.5 + z_g \cdot (\text{GC20} + \text{GC30} \cdot z_g)]\} \quad (4.60)$$

$$I_{\text{gout, ECB}} = I_{\text{gin}} \cdot D \cdot \text{TYPE} \cdot \ln\left(\frac{1.0 + \Delta_{\text{si}}}{1.0 + \Delta_{\text{gate}}}\right) \quad (4.61)$$

$$I_{\text{gout}} = I_{\text{gout, HVB}} + I_{\text{gout, ECB}} \quad (4.62)$$

Note that the mathematical formulation of these equations is slightly different in the code to prevent numerical difficulties. For example, when the argument to the exponential in (4.58) gets large, one can write

$$\ln[1.0 + \Delta_{\text{si}}] = \ln[1.0 + \exp(A_{\text{si}})] = A_{\text{si}} + \exp(-A_{\text{si}}) - \frac{1}{2} \exp(-2A_{\text{si}}) + \dots \approx A_{\text{si}} \quad (4.63)$$

where A_{si} is the argument to the exponential in (4.58). The approximation requires no exponential or logarithm, will not overflow, and becomes exact in double-precision arithmetic for values of A_{si} greater than 34.

4.12 Gate Tunneling Current

$$V_{\text{fb, ov}} = \begin{cases} \text{TYPEP} \cdot E_{\text{g}}, & \text{if } \text{TYPE} \cdot \text{TYPEP} = -1 \\ 0, & \text{otherwise} \end{cases} \quad (4.64)$$

$$x_{\text{g, ov}} = \text{TYPE} \cdot (V_{\text{b, ov}} - V_{\text{fb, ov}}) / \phi_{\text{T}}, \quad (4.65)$$

where $V_{\text{b, ov}}$ is the voltage across the overlap, from bi to gi.

If **SWIGATE** $\neq 0$ and $I_{\text{gov}} + I_{\text{govHVB}} > 0$

$$x_{\text{ov, s0}} = \Phi_{\text{ov}}(x_{\text{g, ov}}, G_{\text{ov, s}}, G_{\text{ov, s}}^2, x_{\text{mrgov, s}}, \xi_{\text{ov, s}}, x_{\text{g1, ov}}) \quad (4.66)$$

$$V_{\text{ov}} = \phi_{\text{T}} \cdot (x_{\text{g, ov}} - x_{\text{ov, s0}}) \quad (4.67)$$

else

$$V_{\text{ov}} = 0.0 \quad (4.68)$$

$$x_{\text{ov, s0}} = 0.0 \quad (4.69)$$

end

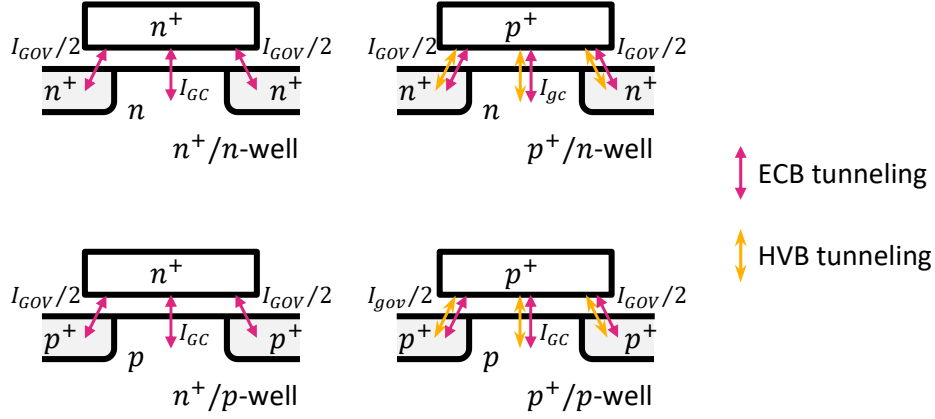


Figure 4.1: Different gate tunneling mechanisms.

Initially,

$$I_{GC} = 0.0 \quad (4.70)$$

$$I_{GOV} = 0.0 \quad (4.71)$$

If **SWIGATE** = true

$$\text{If } I_{gov} + I_{govHVB} > 0 \quad (4.72)$$

$$V_{bov,ig} = V_{b,ov} \cdot \mathbf{TYPE}$$

$$I_{GOV} = I_{gate}(I_{gov}, I_{govHVB}, E_g, V_{ov}, D_{ch}, D_{ch,HVB}, INV_{CHIB}, INV_{CHIB,HVB}, \mathbf{GC2O}, \mathbf{GC3O}, \mathbf{GC2HVO}, \mathbf{GC3HVO}, Q_{CQ}, 0, x_{ov,s0}, \alpha_{b,s}, \alpha_{b,ov}, \phi_T^{-1}, \mathbf{TYPEP}, \mathbf{TYPE}, V_{bov,ig}, B_{OV}, B_{OV,HVB}) \quad (4.73)$$

end

$$\text{If } I_{ginv} + I_{gcHVB} > 0 \quad (4.74)$$

$$V_{bci,ig} = V_C \cdot \mathbf{TYPE}$$

$$V_{ox} = (x_g - x_s) \cdot \phi_T \quad (4.75)$$

$$I_{GC} = I_{gate}(I_{ginv}, I_{gcHVB}, E_g, V_{ox}, D_{ch}, D_{ch,HVB}, INV_{CHIB}, INV_{CHIB,HVB}, \mathbf{GC2O}, \mathbf{GC3O}, \mathbf{GC2HVO}, \mathbf{GC3HVO}, Q_{CQ}, 1, x_s, \alpha_{b,s}, \alpha_{bci,ig}, \phi_T^{-1}, \mathbf{TYPEP}, \mathbf{TYPE}, V_{bci,ig}, B_{CH}, B_{CH,HVB}) \quad (4.76)$$

end

end

Note that the macro I_{gate} is defined in Section 4.11.

4.13 Currents

4.13.1 DC Currents

$$\text{Current through the gate terminal } I_{g, DC} = I_{GC} + I_{GOV}, \quad (4.77)$$

4.13.2 Resistor Currents

If **SWRES** = 1, then the resistor currents are as follows.

Current through Rgsal:

$$I_{Rgsal} = V_{Rgsal} \cdot G_{gsal} \quad (4.78)$$

where V_{Rgsal} is the voltage across the resistance R_{gsal} .

Current through Rgpv:

$$I_{Rgpv} = V_{Rgpv} \cdot G_{gpv} \quad (4.79)$$

where V_{Rgpv} is the voltage across the resistance R_{gpv} .

Current through Rsub:

$$I_{Rsub} = V_{Rsub} \cdot G_{sub} \quad (4.80)$$

where V_{Rend} is the voltage across the resistance R_{end} .

Current through Rend:

$$I_{Rend} = V_{Rend} \cdot G_{end} \quad (4.81)$$

where V_{Rend} is the voltage across the resistance R_{end} .

4.14 Terminal Charges

Total charge at the gate

$$Q_{g, total} = (V_{gb1} - \phi_s - \phi_p) \cdot L_{eff} \cdot W_{eff} \cdot C_{ox, qm} \cdot \mathbf{TYPE} + C_{fr} \cdot V_{C, fr} \quad (4.82)$$

Total charge at the bulk

$$Q_{b, total} = -(V_{gb1} - \phi_s - \phi_p) \cdot L_{eff} \cdot W_{eff} \cdot C_{ox, qm} \cdot \mathbf{TYPE} - C_{fr} \cdot V_{C, fr} \quad (4.83)$$

4.15 Noise

4.15.1 Thermal Noise

Thermal noise contributions from $rgsal$, $rgpv$, $rend$, $rsub$ and $rac0$ are calculated when **SWRES** = 1.

When **SWRES** = 0, the noise contribution is 0. This is realized in the code by setting G_{gsal} , G_{gpv} , G_{end} , G_{sub} and G_{ac0} to 0.

Listed below are the major equations for thermal noise calculation.

$$rgsalnoise = nt0 \cdot G_{gsal} \quad (4.84)$$

$$rgpvnoise = nt0 \cdot G_{gpv} \quad (4.85)$$

$$rendnoise = nt0 \cdot G_{end} \quad (4.86)$$

$$rsubnoise = nt0 \cdot G_{sub} \quad (4.87)$$

If (**RACNOISE** == 0)

$$racnoise = 0 \quad (4.88)$$

else if (**RACNOISE** == 1)

$$racnoise = nt0 \cdot G_{ac0} \quad (4.89)$$

else if (**RACNOISE** == 2)

$$racnoise = nt0 \cdot G_{ac} \quad (4.90)$$

end

4.15.2 Shot Noise

Shot noise contributions from the gate tunneling current are calculated when **SWIGATE** = 1. Below are the equations for shot noise calculation.

$$igcnoise = 2 \cdot q \cdot |I_{GC}| \quad (4.91)$$

$$igovnoise = 2 \cdot q \cdot |I_{GOV}| \quad (4.92)$$

Section 5

Parasitic Elements and Parameter Extraction

5.1 Parasitic Elements

5.1.1 Fringing Capacitance C_{fr}

Overlap and fringing capacitances are lumped together into C_{fr} . Parasitic capacitances along the gate width and length are included. The length component is generally neglected in practice since the poly edge termination along the lengths is over field or shallow trench oxide and the length is small compared to the width. The total fringe capacitance is¹

$$C_{fr} = 2 \cdot (\text{CFRW} \cdot W + \text{CFRL} \cdot L) \quad (5.1)$$

5.1.2 Well and Accumulation Resistance R_{ac} , R_{sub} , and R_{end}

The well or bulk under the gate oxide forms a parallel combination of a bias independent well resistance and a bias dependent accumulation resistance. The assumption of bias independence of the well resistance in depletion follows from the high doping of the surface regions in sub-micron MOSFET technologies; variation of the thin depletion region under the oxide has a negligible effect on the resistance of the well. Measurements validate this assumption; they show no appreciable well resistance variation with gate bias. The bias independent well resistance is

$$R_{sub} = \frac{R_{shs, T} \cdot L}{12 \cdot (W + \text{DWR})} \quad (5.2)$$

As the gate bias increases above the flatband voltage, an accumulation layer forms, therefore

$$R_{ac} = \frac{L}{12 \cdot W \cdot \mu_{acv} \cdot Q_{ac}} \quad (5.3)$$

¹In the MOSVAR model, "m" factor scaling of resistance and charge is implemented directly in the relative current and capacitance equations.

where $\mu_{acv} = \mathbf{UAC} / [(1 + \mathbf{UACRED} \cdot (\mathbf{VFBO} - V_{GBi}))]$

including low-field surface mobility and mobility degradation parameters, and Q_{ac} is the accumulation charge density determined directly from the surface potential.

A significant portion of the MOS varactor resistance is from the well end resistance, formed by the lightly doped source/drain contact regions, the salicided contact diffusion, and the contact resistance. This resistance does not change with gate length and is the factor that limits the maximum quality factor Q for minimum L . The end resistance is given by

$$R_{\text{end}} = \frac{R_{\text{end}, T}}{2 \cdot (W + \mathbf{DWR})} \quad (5.4)$$

where the factor 2 accounts for the symmetry of the gate segment.

5.1.3 Poly Gate Resistance R_{gsal} and R_{gpv}

The salicided poly gate resistance is given by the well known expression

$$R_{\text{gsal}} = \frac{R_{\text{shg}, T} \cdot W}{L \cdot [3 + 9 \cdot (\mathbf{NGCON} - 1)]} \quad (5.5)$$

The silicide to poly contact resistance as described by Litwin [6] and implemented into a MOSFET gate resistance model by Scholten et al. [7], is given by

$$R_{\text{gpv}} = \frac{R_{\text{pv}, T}}{W \cdot L} \quad (5.6)$$

The two gate resistance components enable accurate, scalable modeling of total gate resistance. Additionally, the **gii** node allows for physical placement of the fringing capacitance C_{fr} , directly affecting the quality factor.

5.2 Model Parameter Extraction

The MOS varactor model parameters are physically extracted based on the following set of test devices.

- 1) Varying L (0.5 μm to 5 μm) at fixed $W = 3 \mu\text{m}$, $m = 20$.
- 2) Varying W (3 μm to 16 μm) at fixed $L = 0.5 \mu\text{m}$, $m = 20$.

5.2.1 Capacitance-Related Model Parameter Extraction

Based on the surface potential MOS charge model described, the critical model parameters that describe the first order C-V characteristics are oxide thickness **TOXO**, flatband voltage **VFBO**, and surface well doping **NSUBO**, all of which are closely linked to the process. Since QM effects are included in the charge based model, **TOXO** represents the physical oxide thickness as opposed to an effective or electrical thickness used in non-QM inclusive models.

The MOS varactor capacitance defined in the MOSVAR model is given by

$$C(V) = C_o(V) \cdot L \cdot W + C_{fr} \quad (5.7)$$

where C_o is the bias-dependent capacitance of an intrinsic device, $L = L_g + \text{DLQ}$, and $W = W_g + \text{DWQ}$. The total fringe capacitance is given by

$$C_{fr} = 2 \cdot \text{CFRW} \cdot W + 2 \cdot \text{CFRL} \cdot L \quad (5.8)$$

Parameters DWQ and DLQ describe the deviations of the effective channel length L and width W from their “drawn” values L_g and W_g .

The task is to extract DWQ, DLQ, CFRW and CFRL from the test data. In the MOSVAR model, CFRW encompasses the poly gate overlap of the well contact region (N^+ in the case of NWELL) and the fringe capacitance of the poly edge. A separate term for the overlap region is not included in the model. Previously the parameters DWQ and DLQ were extracted assuming for simplicity that $\text{CFRL} = 0$ [5]. The CFRL term in the MOSVAR model encompasses capacitance associated with the poly gate extension onto STI and fringe capacitances of the poly edge along the length. Neglecting CFRL presents difficulties in obtaining accurate $C(V)$ and $I_g(V)$ extractions over a wide range of device geometries. In this work we present a new technique free of this assumption.

Similar to [5], the improved technique considers C_{min} and C_{max} values of C for the high frequency capacitance, C , extracted from S-parameters with proper de-embedding [8]. To assure identical conditions for measurements and simulations special attention is paid to the frequency behavior of C in the inversion region which directly influences the C_{min} value. Fig. 5.1 displays typical $C(V)$ curve with and without illumination. Without illumination MOS varactor at least partially enters deep depletion regime producing erratic values of C_{min} and complicating comparison with simulated results. Under illumination the time required for the inversion layer formation is reduced and classical high-frequency MOS $C(V)$ curve is experimentally observed. The same $C(V)$ curve is simulated with MOSVAR v1.0.0 model which physically includes the frequency dependence of the inversion layer response to the applied voltage. Thus illumination of the sample is required in order to produce consistent measurements of C_{min} for different devices.

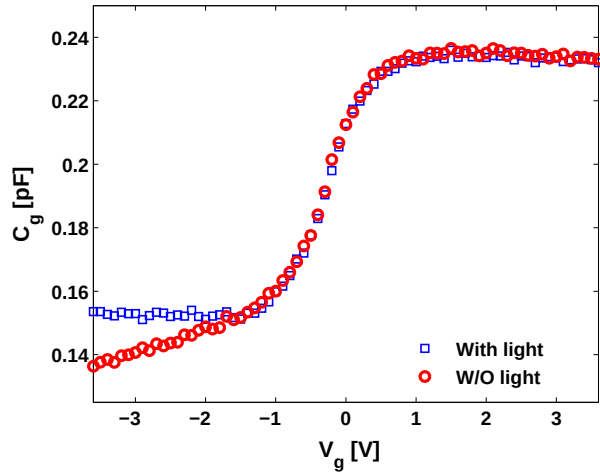


Figure 5.1: Comparison of measured data between light-shedding and dark condition.

The extraction technique relies on regression fitting of C_{max} and C_{min} values over geometry. From (5.7), formally $C_{min}(W_g) = C_{max}(W_g)$ for $W_g = -\text{DWQ}$. Hence the intercept point of $C_{min}(W_g)$ and $C_{max}(W_g)$ lines for fixed L_g (shown in Fig. 5.2) yields:

$$XINT_{wg} = -DWQ \quad (5.9)$$

$$YINT_{cwg} = 2 \cdot CFRL(L_g + DLQ) \cdot m \quad (5.10)$$

Similar extrapolation of the $C_{min}(L_g)$ and $C_{max}(L_g)$ for fixed W_g plots shown in Fig. 5.3 yields

$$XINT_{lg} = -DLQ \quad (5.11)$$

$$YINT_{clg} = 2 \cdot CFRW \cdot (W_g + DWQ) \cdot m. \quad (5.12)$$

The parameters CFRL and CFRW are computed from (5.10) and (5.12), respectively, based on DLQ and DWQ from (5.9) and (5.11). To verify this method, the parameter set thus extracted used to fit the test data for 180nm, 130nm and 65nm technology node. Typical results for different geometries are shown in Figs. 5.4 and 5.5.

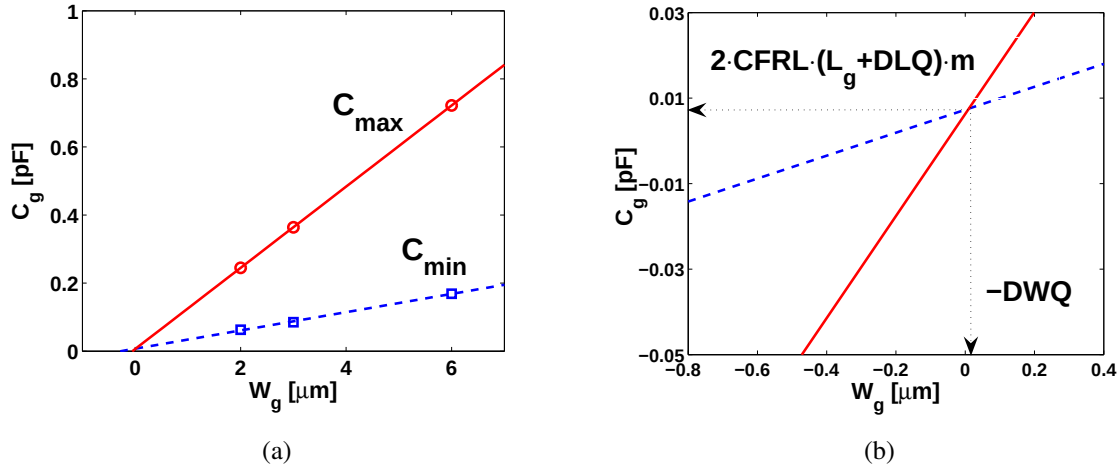


Figure 5.2: (a) Measured $C_{min}(W_g)$ and $C_{max}(W_g)$ of 180nm RFCMOS technology for $L_g = 0.5 \mu m$, $m = 20$ and $f = 500$ MHz and (b) Zoom-in of (a) near intersection point.

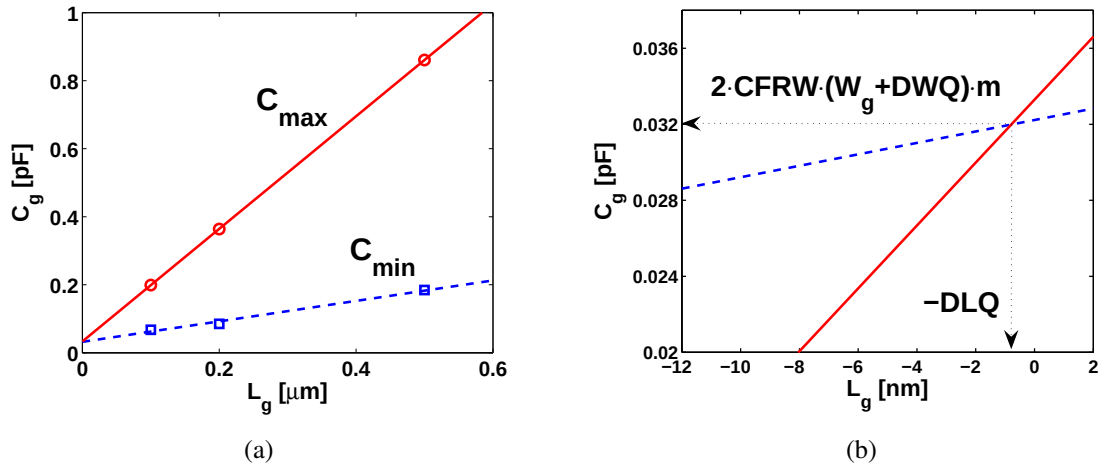


Figure 5.3: (a) Measured $C_{min}(L_g)$ and $C_{max}(L_g)$ of 180nm RFCMOS technology for $W_g = 3 \mu m$, $m = 20$ and $f = 500$ MHz and (b) Zoom-in of (a) near intersection point.

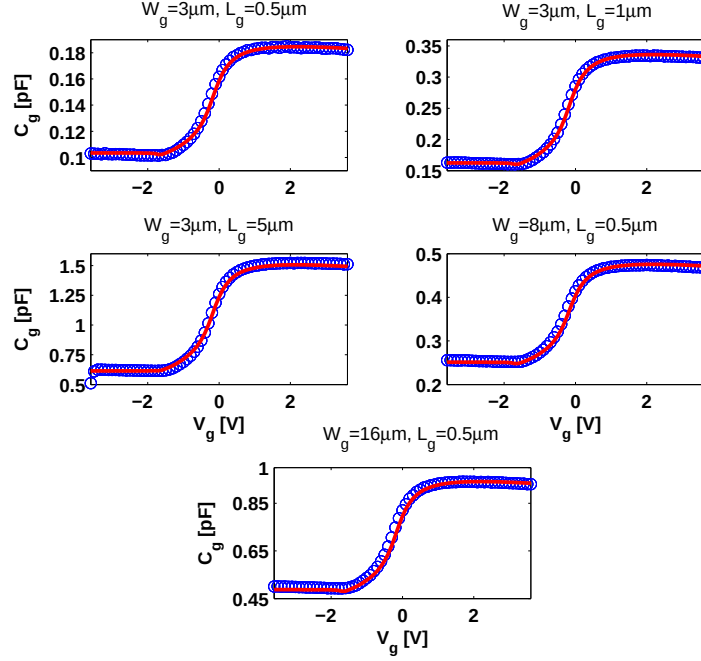


Figure 5.4: $C(V)$ characteristics at $f = 500$ MHz (circles), measured with light-shedding condition, of 180 nm technology node compared with simulation results (solid lines) for different device geometries; $m = 20$.

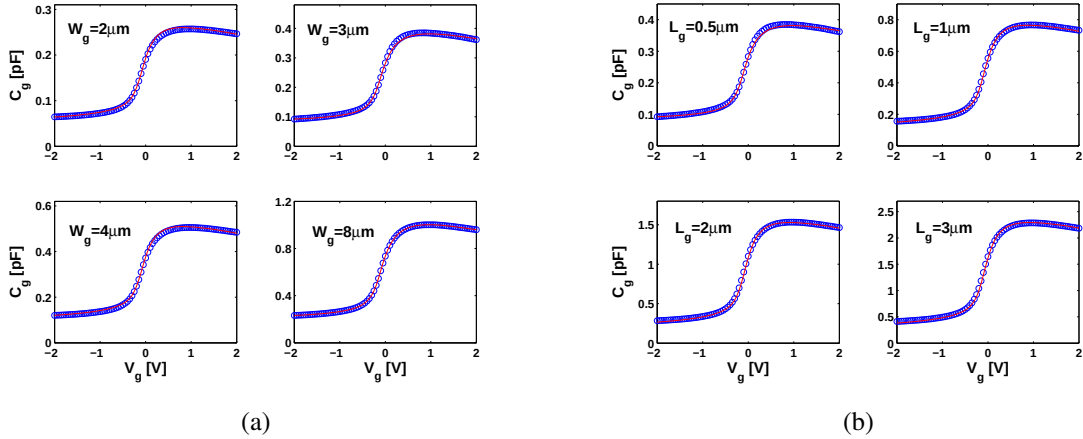


Figure 5.5: (a) Measured (circles) and simulated (solid lines) capacitances ($f = 1$ GHz) of 130 nm technology node for various W_g at $L_g = 0.5 \mu m$ and $m = 20$. (b) Measured (circles) and simulated (solid lines) capacitances ($f = 1$ GHz) of 130 nm technology node for various L_g at $W_g = 3 \mu m$ and $m = 20$.

5.2.2 Resistance-Related Model Parameter Extraction

In order to model the MOS varactor quality factor, also known as its Q characteristics, all resistor associated components should be physically extracted from test devices with varying L and W . Given the high NWELL sheet and end resistance compared to metal and poly resistance, the NWELL resistance dominates the total resistance and hence Q . The metal resistors R_{gm} , R_{sm} and poly-gate resistor R_{gsal} are directly calculated with direct input of the process sheet resistance data ($\rho_{m1} = 66$ m Ω /sq, $\rho_{poly} = 6$ Ω /sq). To extract NWELL

resistance related parameters, the measured data at $V_{GB} = 0$ corresponding to a slight depletion state is used so as to eliminate the effects of accumulation resistance and isolate the NWELL resistance. The extracted NWELL resistance is thus given by

$$R_{nwx} = R_{meas@V_{GB}=0} - (R_{gm} + R_{sm} + R_{gsal}). \quad (5.13)$$

Equating this R_{nwx} to the model derived sum $R_{nwb} + R_{nwe}$, the model parameters $\rho_{nwell} = 314 \Omega/\text{sq}$ and $\rho_{end} = 0.24 \text{ m}\Omega\cdot\text{m}$ are directly extracted for test devices with varying L at fixed W using linear regression fitting as shown in Fig. 5.6(a). The process monitored ρ_{nwell} under thin oxide is 400Ω . Thus the extracted value, which is specific to the MOS varactor, is in line with the process value. Fig. 5.6(b) shows the extracted NWELL resistance plotted for varying $1/W$ at fixed L . As expected, a linear relationship holds.

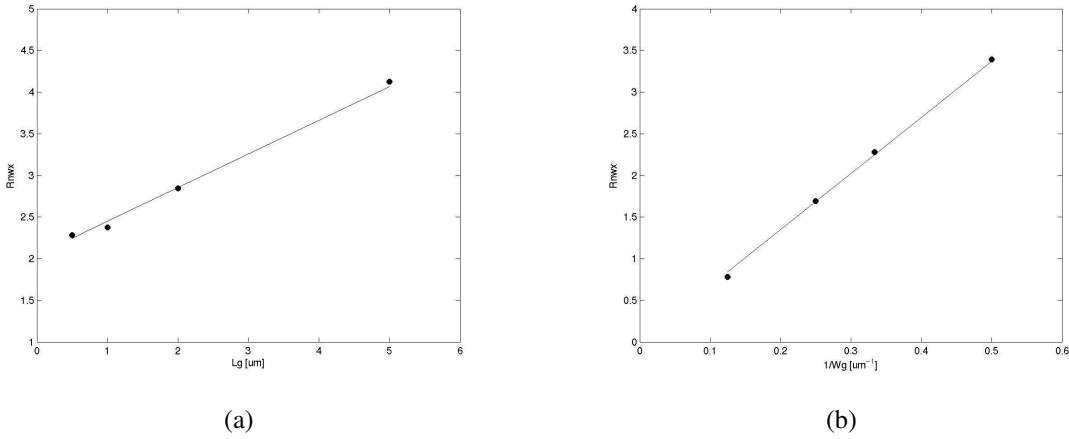


Figure 5.6: (a) Measurement extracted NWELL resistance at $f = 3 \text{ GHz}$, ($V_{GB} = 0 \text{ V}$) versus L compared with model extraction (solid line) at $W = 3 \mu\text{m}$, $m=20$; (b) Measurement extracted NWELL resistance at $f = 3 \text{ GHz}$, ($V_{GB} = 0 \text{ V}$) versus $1/W$ compared with model extraction (solid line) at $L = 0.5 \mu\text{m}$, $m=20$.

5.3 Model Validation

After all the capacitance and resistance related parameters are extracted from physical extraction techniques as previously outlined, the model is simulated against the entire geometry, voltage, and frequency space for C and Q , the critical performances for MOS varactor operation. There are no additional optimization steps performed at this point. Figs. 5.7-5.8 illustrate the MOS varactor model performance in terms of capacitance and over geometry, voltage, and frequency. Fig. 5.7 displays simulated and measured data for varying L at fixed W and m . As L increases, the $C_{\max}/C_{\min}$ ratio increases due to increased oxide capacitance relative to fixed fringing capacitance. The minimum Q decreases as a result of increasing NWELL resistance with increasing L . Fig. 5.8 displays simulated and measured data for varying W at fixed L and m . Q remains relatively constant up to $W = 4 \mu\text{m}$. As W increases to $8 \mu\text{m}$, Q shows a decrease due to higher gate poly resistance.

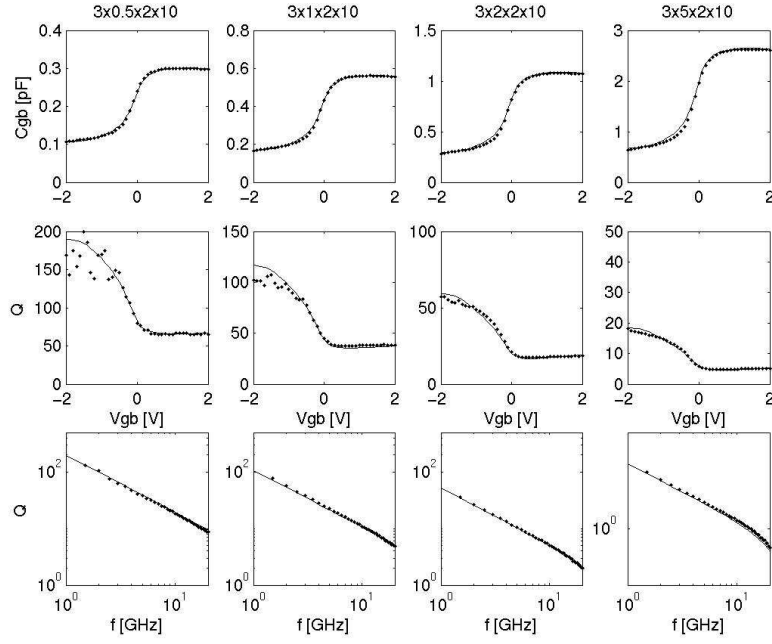


Figure 5.7: Measured $C - V$ (500 MHz), $Q - V$ (3 GHz), and $Q - f$ ($V_{GB} = 1$ V) (solid dots) compared with simulation (solid lines) for varying L at fixed $W = 3 \mu\text{m}$ and $m=20$.

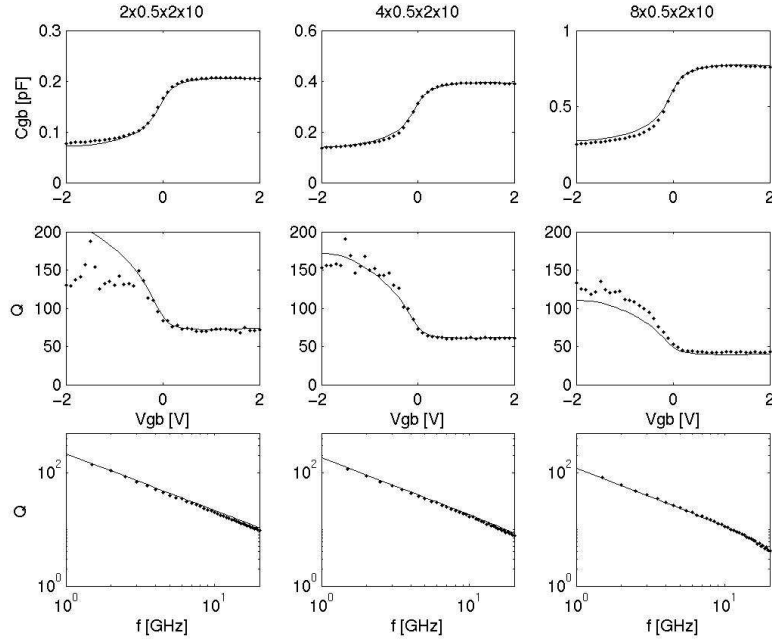


Figure 5.8: Measured $C - V$ (500 MHz), $Q - V$ (3 GHz), and $Q - f$ ($V_{GB} = 1$ V) (solid dots) compared with simulation (solid lines) for varying W at fixed $L = 0.5 \mu\text{m}$ and $m=20$.

Appendix A

Auxiliary Equations

In this Appendix, some auxiliary functions which are used in the model equations are defined.

The MINA-smoothing function:

$$\text{MINA}(x, y, a) = \begin{cases} y - \frac{1}{2} \left(y - x + \sqrt{(y-x)(y-x)+a} \right), & \text{if } (y-x) > \text{'MEPS} \\ y - \frac{1}{2} a / \left(x - y + \sqrt{(x-y)(x-y)+a} \right), & \text{if } (x-y) > \text{'MEPS} \\ y - \frac{1}{2} \left(y - x + \sqrt{\text{'MEPSSQ} + a} \right), & \text{otherwise} \end{cases} \quad (\text{A.1})$$

The MAXA-smoothing function:

$$\text{MAXA}(x, y, a) = \begin{cases} y + \frac{1}{2} \left(x - y + \sqrt{(x-y)(x-y)+a} \right), & \text{if } (x-y) > \text{'MEPS} \\ y + \frac{1}{2} a / \left(y - x + \sqrt{(y-x)(y-x)+a} \right), & \text{if } (y-x) > \text{'MEPS} \\ y + \frac{1}{2} \left(x - y + \sqrt{\text{'MEPSSQ} + a} \right), & \text{otherwise} \end{cases} \quad (\text{A.2})$$

where 'MEPS = 10^{-16} and 'MEPSSQ = 10^{-32} .

σ_1 , and σ_2 , which are used in the explicit approximation of surface potential:

$$\nu = a + c \quad (\text{A.3})$$

$$\mu_1 = \frac{v^2}{\tau} + \frac{c^2}{2} - a \quad (\text{A.4})$$

$$\sigma_1(a, c, \tau, \eta) = \frac{a \cdot \nu}{\mu_1 + (c^2/3 - a) \cdot c \cdot \nu / \mu_1} + \eta \quad (\text{A.5})$$

$$\mu_2 = \frac{v^2}{\tau} + \frac{c^2}{2} - a \cdot b \quad (\text{A.6})$$

$$\sigma_2(a, b, c, \tau, \eta) = \frac{a \cdot \nu}{\mu_2 + (c^2/3 - a \cdot b) \cdot c \cdot \nu / \mu_2} + \eta \quad (\text{A.7})$$

A polynomial function:

$$\text{P3}(u) = 1 + u \cdot [1 + 0.5 \cdot u \cdot (1 + u/3)] \quad (\text{A.8})$$

Three functions are defined to prevent over-/under-flow for exponential functions:

$$\text{expl}(x) = \begin{cases} \exp(x), & \text{if } |x| < k_{\text{se1}} \\ \frac{10^{-100}}{\text{P3}(-k_{\text{se1}} - x)}, & \text{if } x < -k_{\text{se1}} \\ 10^{100} \cdot \text{P3}(x - k_{\text{se1}}), & \text{otherwise} \end{cases} \quad (\text{A.9})$$

$$\text{expl}_{\text{low}}(x) = \begin{cases} \exp(x), & \text{if } x > -k_{\text{se1}} \\ \frac{10^{-100}}{\text{P3}(-k_{\text{se1}} - x)}, & \text{otherwise} \end{cases} \quad (\text{A.10})$$

$$\text{expl}_{\text{high}}(x) = \begin{cases} \exp(x), & \text{if } x < k_{\text{se1}} \\ 10^{100} \cdot \text{P3}(x - k_{\text{se1}}), & \text{otherwise} \end{cases} \quad (\text{A.11})$$

References

- [1] G. Gildenblat, X. Li, W. Wu, H. Wang, A. Jha, R. van Langevelde, G.D.J. Smit, A.J. Scholten, and D.B.M. Klaassen, "PSP: An advanced surface-potential-based MOSFET model for circuit simulation," *IEEE Trans. Electron Devices*, vol. 53, no. 9, pp. 1979-2003, Sept. 2006.
- [2] J. Victory, Z. Zhu, Q. Zhou, W. Wu, G. Gildenblat, Z. Yan, J. Cordovez, C. McAndrew, F. Anderson, J.C.J. Paasschens, R. van Langevelde, P. Kolev, R. Cherne, and C. Yao, "PSP-based scalable MOS varactor model," *IEEE Custom Integrated Circuits Conference*, pp. 495-502, 2007
- [3] <https://si2.org/cmc-standard-models/#other>
- [4] J. Victory, C. McAndrew, and K. Gullapalli, "A time-dependent, surface potential based compact model for MOS capacitors," *IEEE Electron Device Lett.*, vol. 22, no. 5, pp. 245-247, May 2001.
- [5] J. Victory, Z. Yan, G. Gildenblat, C. McAndrew, and J. Zheng, "A physically based, scalable MOS varactor model and extration methodology for RF applications," *IEEE Trans. Electron Devices*, vol. 52, no. 7, pp. 1343-1353, Jul. 2005.
- [6] A. Litwin, "Overlooked interfacial silicidepolysilicon gate resistance in MOS transistors," *IEEE Trans. Electron Devices*, vol. 48, no. 12, pp. 2179-2181, Dec. 2001.
- [7] A. J. Scholten, L. F. Tiemeijer, R. van Langevelde, R. J. Havens, A. T. A. Zegera-van Duijnhoven, and V. C. Venezia, "Noise modeling for RF CMOS circuit simulation," *IEEE Trans. Electron Devices*, vol. 50, no. 5, pp. 618-632, May 2001.
- [8] Z. Zhu, J. Victory, S. Chaudhry, L. Dong, Z. Yan, J. Zheng, W. Wu, X. Li, Q. Zhou, P. Kolev, C. McAndrew, and G. Gildenblat, "Improved parameter extraction procedure for PSP-based MOS varactor model," in *Proc. IEEE Int. Conf. Microelectronic Test Structures (ICMTS'09)*, pp. 148-153, 2009